



⚡ VoltTrack Platform

VoltTrack — EV Telemetry & Driver Monitoring System

A Full-Stack Fleet Management & Analytics Platform for Electric Vehicles

Project Presentation | Full-Stack & Machine Learning Capstone

Prepared for Project & Portfolio Review • Date: August 2026

The Challenge in Modern EV Fleet Operations

The Operational Challenge

- **Data Fragmentation:** EV fleet operators manage thousands of telemetry signals, but battery health, vehicle location, driver behavior, and charging status are scattered across disconnected systems.
- **Invisible Downtime Risks:** Without continuous monitoring of State of Charge (SoC) and State of Health (SoH), unexpected range anxiety and unpredicted battery degradation lead to severe operational delays.
- **Lack of Driver Visibility:** Drivers lack a personalized cockpit view to track live range estimates, trip efficiency ratings, eco-scores, and nearby compatible charging stations.

Why It Matters Now

- **Rapid EV Adoption:** Commercial fleets are rapidly transitioning to electric powertrains, requiring specialized fleet management software built specifically for EV metrics.
- **High Operating Expenses:** Unplanned maintenance and inefficient charging schedules increase total cost of ownership (TCO) significantly.
- **Driver Safety & Longevity:** Monitoring driving behavior directly impacts battery life cycle, vehicle safety, and insurance liabilities.
- **Sustainability Tracking:** Fleet operators require accurate CO2 reduction measurements and energy consumption reporting for ESG compliance.

Core Platform Goals & Scope



Role-Based Access Control

Build an enterprise portal with separate Admin and Driver workflows. Admins monitor overall fleet KPIs, drivers, and revenue while drivers get a focused vehicle cockpit view.



Real-Time EV Telemetry

Track continuous telemetry parameters including State of Charge (SoC), State of Health (SoH), live GPS location, speed, odometer readings, and maintenance due alerts across 15,000+ vehicles.



Driver Cockpit Experience

Empower drivers with a personalized cockpit view featuring live range estimations, interactive maps, nearby charging station locations via Open Charge Map API, and trip PDF downloads.



ML Range Prediction Model

Implement a machine learning regression pipeline to accurately predict remaining battery range based on real-time telemetry inputs, vehicle specifications, and driving environmental factors.

Full-Stack Technology Architecture



Frontend UI

- **React (Vite)**
High-performance SPA framework
- **Tailwind CSS**
Custom VoltTrack design system
- **Recharts**
Interactive analytical charts
- **Leaflet Maps**
Live GIS map integration
- **Lucide React**
Clean modern vector icon set



Backend API

- **Node.js**
Asynchronous JavaScript runtime
- **Express.js 5**
RESTful API routes & middleware
- **Prisma ORM**
Type-safe database mapping
- **Bcrypt Hashing**
Secure salted password security
- **JWT / Sessions**
Protected route authentication



Database

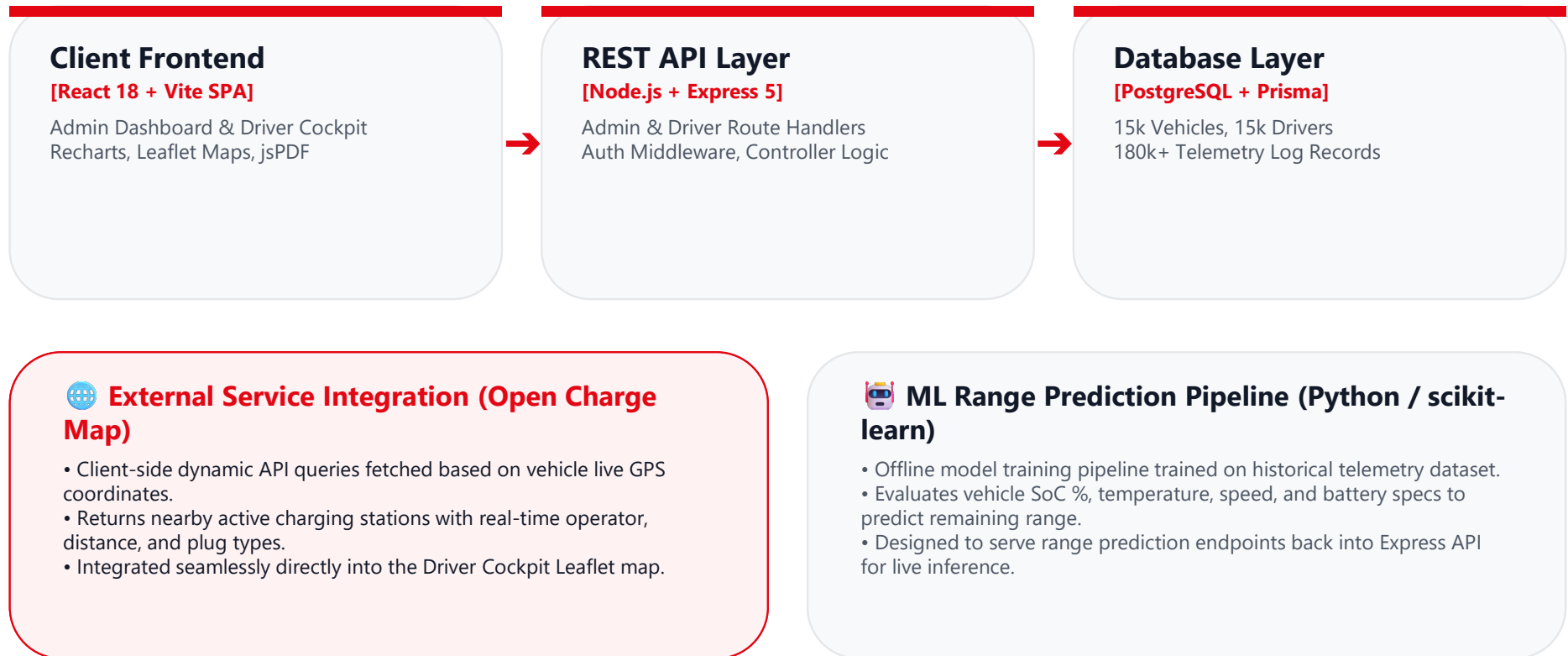
- **PostgreSQL**
Enterprise relational database
- **Prisma Pg Adapter**
Optimized connection pool
- **15k+ Vehicles**
Complete vehicle spec data
- **15k+ Drivers**
Full identity & license records
- **180k+ Telemetry**
Monthly sensor log dataset



ML & External

- **Python & Pandas**
Data preprocessing & feature eng.
- **scikit-learn**
Regression modeling algorithms
- **Linear & Forest**
Evaluated multiple ML models
- **Open Charge Map**
Live external API for chargers
- **jsPDF & AutoTable**
Client-side PDF report generator

End-to-End System Components & Data Flow



Relational Schema & Enterprise Scale

Core Database Entities (Prisma Schema)

- **Table: admins**
Fields: id, email, password_hash, full_name, role, created_at
- **Table: drivers**
Fields: driver_id, full_name, email, phone, license_number, status
- **Table: vehicles**
Fields: vin, model, manufacturer, battery_capacity_kwh, torque_nm, dimensions
- **Table: telemetry_records**
Fields: id, vehicle_id, driver_id, soc_percent, soh_percent, range_km, speed, location

15,000

Registered Fleet Vehicles

Complete specs: dimensions, torque, battery kWh & specs

15,000

Active Fleet Drivers

Real driver profile data: license #, phone & email contact info

180,000+

Monthly Telemetry Records

Continuous sensor logs: SoC %, SoH %, speed & GPS coordinates

Multi-Tiered Access & Protection Standards



Dual Portal Role-Based Access

Separate login portals for Admins and Drivers ensure complete security isolation. Admins gain access to fleet-wide controls, while drivers only view their assigned vehicle telemetry.



Bcrypt Password Hashing

All user credentials (admins and drivers) are hashed using bcrypt with salt rounds before database insertion. Plaintext passwords are never logged or stored.



Auto-Generated Unique IDs

Smart ID generators ensure sequential pattern compliance for new registrants — auto-assigning IDs like ADM-1004 for admins and DRV-15024 for drivers seamlessly.



OTP Password Reset & Guard

Includes a complete OTP-based password recovery flow with countdown timers. Admin account creation is restricted exclusively to authenticated existing admins.

Fleet-Wide Real-Time Analytics & Control

■ Fleet-Wide KPI Cards

Live counts for Total Vehicles, Active Working fleet, Vehicles in Garage for maintenance, and Vehicles currently Charging.

■ Analytical Charts

Interactive Recharts visualizer for Revenue Leaderboard, Maintenance Status Breakdown, and Monthly Electricity Consumption.

■ Monthly Insights

Automated analytical highlights identifying the Highest Revenue Driver, Top Maintenance Manufacturer, and Peak Energy Usage.

■ Vehicle Management Table

Searchable and filterable table displaying real-time SoC %, SoH %, maintenance status, and quick admin action triggers.

Personalized EV Vehicle Cockpit

Live Vehicle Identity & Telemetry

Real-time battery State of Charge (SoC %), estimated range (km), battery health (SoH %), and current vehicle status.

Interactive Map & Charging Stations

Live vehicle location mapped on Leaflet GIS with nearby charging stations fetched dynamically from Open Charge Map API.

Eco-Score & Sustainability

Personalized driving efficiency score and calculated monthly CO2 emissions savings.

Trip History & PDF Downloads

Complete log of recent trips with distance, energy used, and downloadable client-side PDF trip reports.

Proactive Alert Banner

Instant warning banners triggered for low battery levels (<20%) or upcoming maintenance requirements.

Predictive Battery Range Estimation Model



Objective: Predict EV Remaining Range (km) from Real-Time Telemetry & Specs

Input Features: Battery SoC %, State of Health (SoH %), Speed (km/h), Temperature, Battery Capacity (kWh), & Vehicle Weight.

Model Name	Algorithm Type	R ² Score (Accuracy)	Performance Evaluation
Linear Regression	Standard OLS Linear Model	~0.85	Fast baseline, struggles with non-linear factors
Decision Tree Regressor	Non-Linear Decision Tree	~0.91	Captures non-linear feature interactions well
Random Forest Regressor ★	Ensemble Tree Model	~0.96	Highest precision & lowest error across test set

** Note: Model trained in Python using scikit-learn. R² metrics indicative of feature correlation runs in Jupyter notebook.*

UI Screenshots & Live Demo Placeholders



Admin Dashboard View

Insert Screenshot:
Admin Analytics, KPIs, Recharts & Fleet Inventory Table



Driver Cockpit View

Insert Screenshot:
Live Telemetry, SoC %, Range & Dynamic Alerts



Dual Portal Auth & OTP

Insert Screenshot:
Admin / Driver Login & Password Reset Flow



Interactive Map & Charging

Insert Screenshot:
Leaflet Map with Open Charge Map Live Stations

Project Summary & Future Roadmap

Project Summary

- **Full-Stack Execution:** Successfully developed an end-to-end EV fleet telematics software application from DB schema to interactive frontend.
- **Real Data Scale:** Engineered relational PostgreSQL schema handling 15k vehicles, 15k drivers, and 180k telemetry records seamlessly.
- **User-Centric Portals:** Delivered specialized, secure portals for both Fleet Admins and EV Drivers with live map and charging integration.
- **Data-Driven Intelligence:** Integrated machine learning prediction models to solve real-world EV range anxiety.

Future Enhancement Roadmap

- **Live SMS Gateway:** Integrate Twilio / SMS gateway for real-time mobile OTP delivery during auth and emergency battery alerts.
- **IoT Telematics Stream:** Connect live MQTT / WebSockets streaming data pipelines for real-time vehicle telematics ingest directly from vehicle CAN bus.
- **Native Mobile Application:** Develop iOS & Android native mobile apps using React Native for driver cockpit access on smartphones.
- **Live ML API Microservice:** Deploy the trained scikit-learn Random Forest model as a FastAPI microservice for real-time inference during active routes.